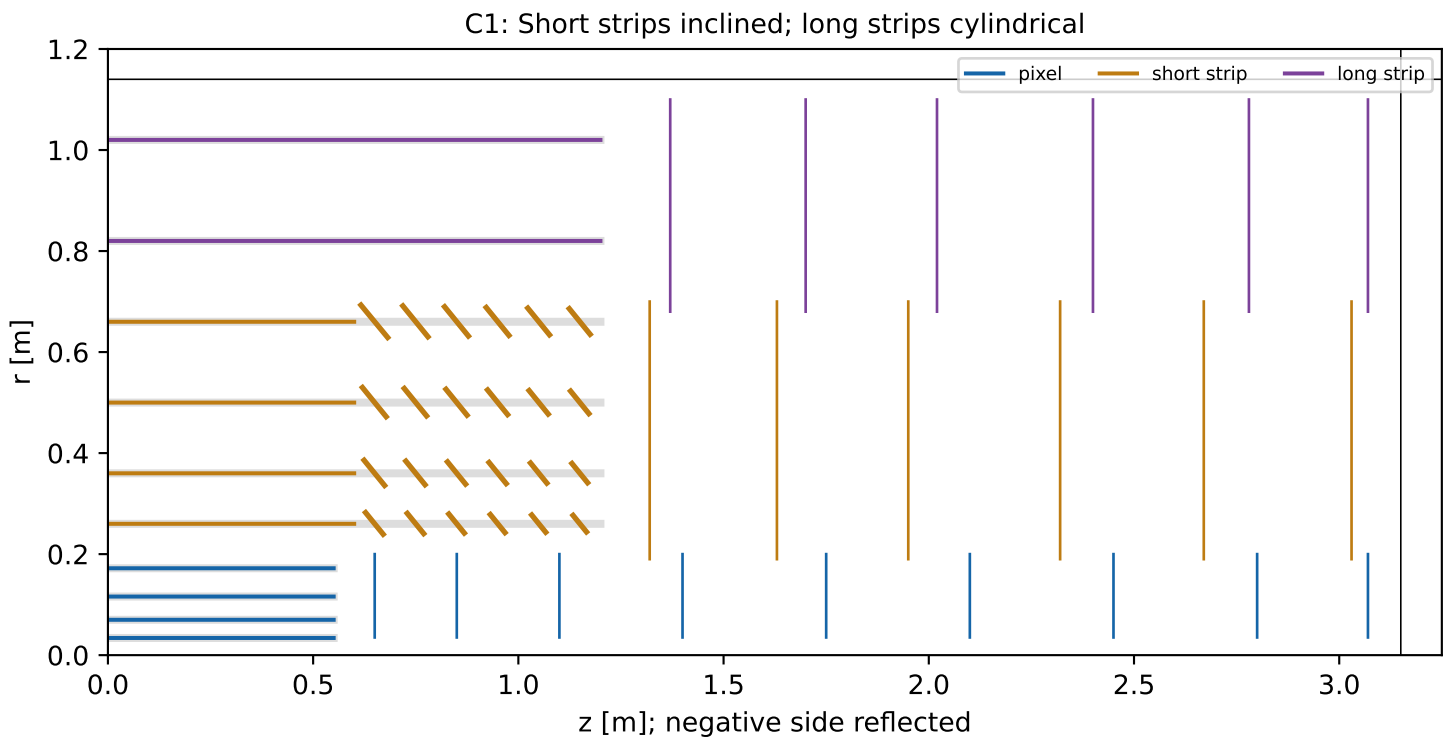


DES-006 C1: Short strips inclined; long strips cylindrical

DRAFT / PROTOTYPE | 2026-09-18 | PR #13 follow-up

C1 is the simpler inclined alternative: keep both outer long-strip barrels cylindrical and incline only the four short-strip (strixel) barrel ends.

All pixels and endcap disks retain A. Tilted barrel sections keep the original station radii and use six rows per end at $|z|=0.65$ to 1.15 m in 0.10 m steps. The central $|z|\leq 0.60$ m section remains cylindrical. Normals aim toward the origin, capped at 45 degrees; angular edges have 10 mm tangent extensions.



The drawing shows conical envelopes of planar module rows, not curved silicon or a tiled detector. Overlapping rows contribute their full local material. Phi gaps, actual module dimensions, supports, services and mechanical clearances remain unresolved.

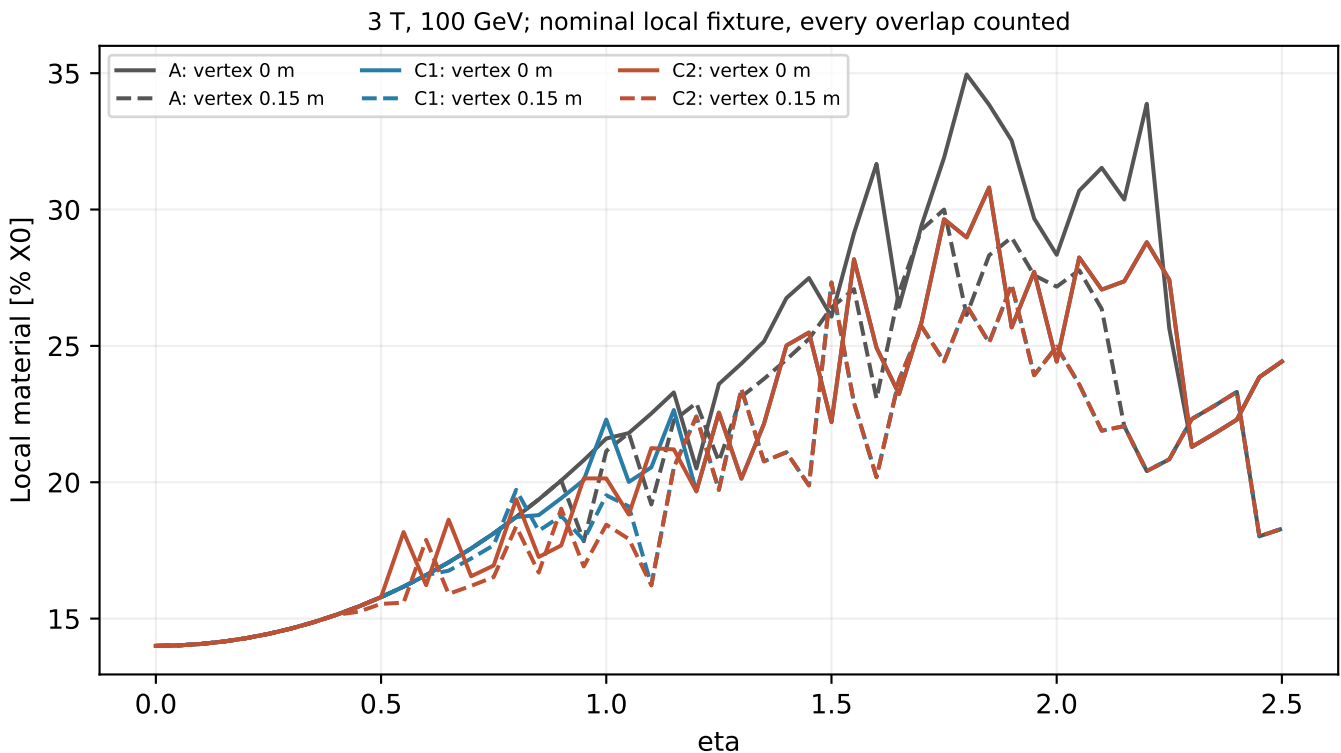
Long strips remain double-sided in every option: two scalar faces per paired station, both faces in silicon area, one 2% X0 allowance per pair crossing. The origin/ $\eta=0$ reference has 10 stations, 12 silicon-face crossings and 20 ideal scalar coordinates.

C1: tradeoffs and optimisation status

Full-system ideal strixel area is 39.939 m² versus A 43.953 m² (-9.13%). Long-strip silicon area stays 111.861 m², counting both faces. C1 has 48 inclined row envelopes and none in long strips. The sampled local-material ratio C1/A spans 0.779 to 1.117.

Both alternatives lose no parent station groups relative to A over 21,627 finite-field probes and a finer zero-field vertex scan. These are sampled coverage checks, not efficiency or continuous coverage proof. Area is an ideal proxy, not installed module cost. No C1/C2 IdRes or material-aware fit result is claimed.

No systematic radial or axial position optimisation has been performed. Barrel radii are inherited ODD controls; disk positions, annuli and barrel lengths are hand-chosen starting hypotheses. Field/material/vertex sweeps and 0/5/10 mm row-margin checks test feasibility and sensitivity, not an optimum. C1/C2 isolate tilt scope without moving the reference radii or row centres.



Recommendation: investigate C1 first for its simpler outer barrels; retain C2 to quantify whether its small extra area saving can justify stereo support/routing complexity. Review the tradeoff before selecting a layout.